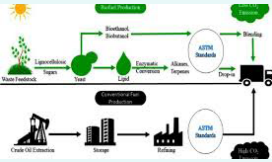


Formal and System Design

Futuristic Systems

1 Biofuel

Biofuel is a **renewable energy** made from organic waste or plants, used to power vehicles and **reduce reliance on fossil fuels**.



2 VTOL

A **vertical takeoff and landing (VTOL)** plane ascends and lands without a runway, allowing for **compact, flexible transport** in urban and remote areas.



3 PV (photovoltaic) Panel Farm

A PV panel farm is a **large-scale solar system** that converts sunlight into **clean electricity** for buildings and communities.



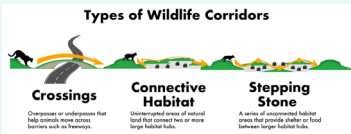
4 Hyperloop

The Hyperloop is a proposed high-speed system where **pressurized pods** travel through **low-pressure tubes** using magnetic levitation for **near-frictionless, ultra-fast** city-to-city travel.



5 Wildlife Corridor

A wildlife corridor is a **natural pathway** that lets animals move safely through urban areas, **supporting biodiversity and reconnecting ecosystems**.



Final Design Solution

Formal Response

This airport proposal is inspired by plants, **biomimicry**, and **swarm behavior**—functioning like a living system that absorbs waste as energy, recycles resources, and adapts through **organic forms and intelligent, nature-driven movement**.



System Response

This future airport operates as a living system, inspired by plants, biomimicry, and swarm behavior. It **absorbs trash as biofuel, recycles water, and harvests solar energy through PV panel farms to power terminals, vertical takeoff aircraft, and commercial spaces**. Its branching form mimics plant growth, while wildlife corridors and reclaimed landfills support surrounding ecosystems. **Swarm-based algorithms guide passenger flow and autonomous vehicles**, creating a responsive, efficient, and ecologically integrated transportation hub for the year 2100.

Design Concept

The airport is conceived as a **living plant**—a **regenerative organism** rooted in reclaimed land. Like a plant draws nutrients, filters water, and powers growth, the airport **absorbs trash as fuel, recycles water, and generates clean energy to support vertical aircraft, terminals, and shops**. This metaphor not only drives its sustainable systems but also inspires the airport's form, with **branching circulation, layered canopies, and adaptive, organic geometries** that echo the logic of **living structures**.

